Azraq Monte Carlo — complete guide and API reference (single file)

This is the one Markdown document for the product: **HTTP vs engine v0/v1/v2**, **integration**, **information flow and validation**, **all 21 routes** (auth, models, errors, curl examples), **real data / calibration**, and **client FAQ**.

Live: Swagger UI · `https://monte-carlo.octaloop.dev/openapi.json` · local `http://127.0.0.1:8000`

PDF / document revision: 1.1.0 (2026-05-05). This tracks **changes to this reference export** — it is **not** the `/v1/` REST prefix, and **not** the optional `semver` query default (`1.0.0`) on `POST /v1/shockpack/catalog/register`.

Print PDF: from repo root run `python scripts/build_api_reference_pdf.py` (writes `docs/Azraq_Monte_Carlo_API_Reference.pdf` and a copy `API_Reference.pdf` at repo root). For extra schema narrative and examples beyond OpenAPI, see **API.md**. A route-only catalogue lives in **ENDPOINTS.md** if you prefer a shorter list.

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Part I — Product overview: v0 / v1 / v2, real data, FAQ

One document for **sales, integration, and validation**. It ties together **what "v0 / v1 / v2" mean in this product**, **which HTTP routes you call**, **how live or file-based data feeds the model**, and **typical client questions**.

I.1 The two "versions" (do not confuse them)

Meaning	What it is	Examples
HTTP API prefix	Every business route today lives under `/v1/...` (plus `/ws/v1/...` for one WebSocket).	`POST /v1/simulate/asset`
Engine / model label	Strings in **response metadata** such as `model_version` and `execution_mode`. These describe *which code path ran*, not the URL prefix.	`model_version: "azraq-mc-v1"`, `execution_mode: "v0_base"`

Important: The deterministic "v0" baseline is **not** mounted at `/v0/...`. It is exposed as `POST /v1/simulate/v0/base-case` — the path is still `/v1`, and the `v0` in the path marks the *baseline* operation.

There is **no** separate public `/v2/...` REST prefix. Portfolio "v2" is the `model_version` string for the **joint multi-asset** simulator, still called via `/v1/simulate/portfolio` (or the WebSocket below).

I.2 Engine v0 — deterministic baseline (spreadsheet parity)

Purpose: One scenario, **no Monte Carlo randomness**. Shock vector is zero; you check that the server's **DSCR / IRR / cashflow-style outputs** match your Excel or internal model for the same `asset` JSON.

Field	Typical value
`model_version`	`azraq-mc-v0`
`execution_mode`	`v0_base`
HTTP operation	`POST /v1/simulate/v0/base-case`
Request body	`asset` only (`AssetAssumptions`) — same shape you use for MC
Response	`BaseCaseResult` : `metadata` + `base` (point estimates: `dscr` , `irr_annual` , `annual_revenue` , `ebitda` , etc.)

Supporting APIs (optional around v0): Any `GET /v1/market/...` or `POST /v1/calibration/preview` are *not required* for v0; they matter when you later build `shockpack` with `dynamic_margins` .

I.3 Engine v1 — single-asset Monte Carlo (main product path)

Purpose: Many correlated random draws → **distributions** of DSCR, covenant breach probability, percentiles, optional attribution, etc.

Field	Typical value
`model_version`	`azraq-mc-v1`
`execution_mode`	`adhoc_asset` (interactive POST) or `scheduled_monitoring` (scheduled POST)
Primary HTTP operations	`POST /v1/simulate/asset` (main), `POST /v1/simulate/scheduled/asset` (same maths + optional on-disk JSON snapshot)

What you send: `asset` + `shockpack` (`ShockPackSpec`), *or* `shockpack_catalog_entry_id` (with optional inline patch), plus optional flags (`performance_profile` , `include_attribution` , full-stack options on `asset` , etc.). Full schema: OpenAPI `/openapi.json` or **API.md**.

I.4 Engine v2 — portfolio (joint shocks across ≥2 assets)

Purpose: One shared random world: scenario *k* uses the **same** shock draw for every asset in the list, so correlations across projects are meaningful.

Field	Typical value
`model_version`	`azraq-mc-v2-portfolio`
`execution_mode`	`portfolio_joint`
HTTP operation	`POST /v1/simulate/portfolio`
Real-time operation	`WebSocket` `WS /ws/v1/simulate/portfolio` (same joint simulation; can stream **progress**)

Requirement: At least two assets with **distinct** `asset_id` values.

I.5 Which API for which job (quick map)

Client need	Call
"Does your math match our spreadsheet?"	`POST /v1/simulate/v0/base-case`
"What's our tail risk / breach % for one project?"	`POST /v1/simulate/asset`
"Run again every week and store JSON on disk."	`POST /v1/simulate/scheduled/asset`
"Two or more projects, same shock world."	`POST /v1/simulate/portfolio` or `WS /ws/v1/simulate/portfolio`
"Prove Yahoo / our file filled volatilities."	`POST /v1/calibration/preview`, then check `calibration_trace` / run `metadata.margin_calibration_trace`
"Inspect copper / VIX history before we trust a symbol."	`GET /v1/market/yfinance/{symbol}/history`
"Reuse a signed-off shockpack."	`POST /v1/shockpack/catalog/register`, then simulate with `shockpack_catalog_entry_id`
"Compare two saved runs."	`POST /v1/snapshots/diff/asset-metrics`

The **numbered route inventory** and **per-endpoint reference** are in Part IV below.

I.6 Real data vs the Monte Carlo core (plain language)

The Monte Carlo core does not need a live market feed to run. It needs:

1. `asset` — *your* deal assumptions (no randomness).
2. `shockpack` — *how* randomness enters: scenario count, seed, `margins` (volatilities / dispersion), **correlation**, copula, optional time structure.

Where "real data" enters: It **calibrates** those margins (and can be audited), via `dynamic_margins` on the shockpack:

Source	Mechanism	Typical use
Yahoo Finance	`dynamic_margins.yahoo_finance` → annualised log vol from closes; `GET /v1/market/yfinance/...` for manual inspection	Commodities, equity indices, some proxies
ECB	`GET /v1/market/ecb/eur-usd` reads public ECB daily XML	FX spot sanity / storytelling (MC still uses your `shockpack` params)
Your HTTP API	`dynamic_margins.http` — trusted URL returns JSON margins	Internal ETL, SOFR/SONIA proxies you host
File	`dynamic_margins.file` — JSON or Excel with margin columns	Signed-off vol sheets from risk team

Order of application (product rule): `file` → `http` → each `yahoo_finance` binding (later steps can override the same margin keys). After materialisation, resolved numbers sit in `margins` and `dynamic_margins` is cleared on the resolved spec.

Proof for auditors / clients: Use `POST /v1/calibration/preview` to see `resolved_shockpack` without paying for MC; on a full run, inspect `metadata.margin_calibration_trace`. The machine-readable list of stress-source ideas is `GET /v1/calibration/stress-data-catalog`.

Security note: File paths and HTTP URLs run **inside the API process**. In production, restrict to **trusted** paths and URLs (SSRF / file-read risk).

I.7 Recommended validation order (before trusting tails)

1. Build `asset` from your systems.
2. Build `shockpack` (start with manual `margins` if needed).
3. Optional: `POST /v1/calibration/preview` if using `dynamic_margins`.
4. `POST /v1/simulate/v0/base-case` — must align with your spreadsheet.
5. `POST /v1/simulate/asset` — interpret `SimulationResult.metrics` (and optional attribution as **indicative** narrative, not a regulatory sign-off by itself).

More detail: Part III.

I.8 Client questions (FAQ)

Q: Why is everything `/v1` if you talk about v0 and v2?
A: `/v1` is the **REST version of the HTTP contract**. **v0 / v1 / v2** in conversation usually mean `model_version` / `execution_mode` (deterministic baseline vs single-asset MC vs joint portfolio).

Q: Is v0 deprecated?
A: No. It is the **recommended sanity check** before Monte Carlo.

Q: Do you stream live prices into the simulation?
A: Not as a continuous feed. **Live (or file) data can refresh volatility parameters** via `dynamic_margins`; the simulation itself draws random scenarios from the **parameterised** shock model you define.

Q: Will Yahoo Finance always work?
A: It depends on `yfinance` and Yahoo's availability. For production calibration, many clients prefer `http` or `file` from their own curated data.

Q: What does reproducibility mean?
A: Fix `asset`, `shockpack.seed`, `shockpack.n_scenarios`, and the resolved `margins`. The same inputs should yield the same `SimulationResult` (subject to documented floating-point / dependency behaviour).

Q: Is the service stateless?
A: Each `POST /v1/simulate/asset` is independent unless **you** choose catalogue ids or snapshot endpoints to reuse stored artefacts.

Q: How do we explain "shocks" to a non-quant?
A: Shocks are **random draws** that scale risk factors (revenue, capex, opex, rate, etc.) according to **volatility and correlation** you set — not a list of hand-picked "−20% events." See Part III §1.

Q: What is full-stack mode?
A: Optional **12-factor** shock design aligned with `GET /v1/catalog/full-stack-factors` and `asset.full_stack.enabled`. Use only when your `shockpack` matches that dimensionality.

Q: What headers might we need?
A: If `AZRAQ_API_KEY` is set: `X-API-Key` on all routes except `GET /health`. Optional: `X-Azraq-User-Id` (audit

label), `X-Azraq-Tenant-Id` (catalogue), `X-Azraq-Catalog-Role` (promote). Details: Part IV §1.

Q: Where do "layer versions" appear?
A: Audit metadata can bundle semantic versions for shock/transform/metrics/impact layers (`layer_versions` in code). These support **incremental recomputation and traceability**, not separate HTTP paths.

I.9 Hosted and local URLs

Environment	Base	Docs
Hosted (example)	`https://monte-carlo.octaloop.dev`	`/docs`, `/openapi.json`
Local dev	`http://127.0.0.1:8000`	same

Part II — Integration quick guide

Short reference for plugging **your data** into the Monte Carlo API. **Every route** is also documented in Part IV. Narrative + extra examples: **API.md**.

II.1 Base URL

Use your deployed host, e.g. `https://api.example.com` or local `http://127.0.0.1:8000`.

II.2 Primary call (single project)

Method / path	`POST /v1/simulate/asset`
Body	JSON `AdhocSimulationRequest`: at minimum `asset` + `shockpack`
Response	`SimulationResult`: `metadata`, `metrics`, optional `attribution`, etc.

Health check (no API key on typical setups): `GET /health`

II.3 Authentication & headers

Env / config	Client header
If `AZRAQ_API_KEY` is set on the server	Send `X-API-Key: <same value>` on requests that require it (`GET /health` is usually public).

Optional audit labels (not login):

- `X-Azraq-User-Id`
- `X-Azraq-Tenant-Id` (used for some catalogue routes)

II.4 "Reset" and state

The service is **stateless**: each `POST /v1/simulate/asset` is independent. There is **no server-side session** to clear. To "reset," **discard the previous response** and **POST new JSON**.

To **persist** full JSON snapshots on the server for later diffing, see `POST /v1/simulate/scheduled/asset` in Part IV.

II.5 Minimal request shape

```
{
  "shockpack": {
    "shockpack_id": "your-pack-id",
    "seed": 42,
    "n_scenarios": 2500,
    "sampling_method": "monte_carlo",
    "margins": {
      "revenue_log_mean": 0,
      "revenue_log_sigma": 0.08,
      "capex_log_mean": 0,
      "capex_log_sigma": 0.06,
      "opex_log_mean": 0,
      "opex_log_sigma": 0.05,
      "rate_shock_sigma": 0.005
    }
  },
  "asset": {
    "asset_id": "your-asset-id",
    "assumption_set_id": "your-label",
    "horizon_years": 15,
    "base_revenue_annual": 42000000,
    "base_opex_annual": 18000000,
    "utility_opex_annual": 0,
    "initial_capex": 280000000,
    "equity_fraction": 0.35,
    "tax_rate": 0,
    "financing": {
      "debt_principal": 182000000,
      "interest_rate_annual": 0.065,
      "loan_term_years": 18,
      "covenant_dscr": 1.2
    }
  },
  "include_attribution": false,
  "performance_profile": "interactive"
}
```

`performance_profile:` `interactive` caps scenario count for responsiveness; `standard` / `deep` allow larger `n_scenarios` (see OpenAPI / **API.md**).

II.6 What to read from the response

Your product need	Typical field under `metrics`
DSCR distribution	`dscr` (p05, p50, p95, …)
IRR distribution	`irr_annual`
P(covenant breach)	`covenant_breach_probability`
Tail / VaR-style IRR	`var_irr_95`, `cvar_irr_95` (when populated)
Build cost distribution	`total_capex` (when present)

Exact shapes are in **OpenAPI** (`/openapi.json`).

II.7 Mapping checklist (your data → API)

Your field / concept	API location
Project / case id	`asset.asset_id`, `asset.assumption_set_id`
Model horizon	`asset.horizon_years`
Revenue, opex, capex	`asset.base_revenue_annual`, `base_opex_annual`, `initial_capex`
Equity share	`asset.equity_fraction` (decimal, e.g. `0.35`)
Debt, coupon, term, DSCR floor	`asset.financing`: `debt_principal`, `interest_rate_annual` (decimal), `loan_term_years`, `covenant_dscr`
Shock count / reproducibility	`shockpack.n_scenarios`, `shockpack.seed`
Volatility / margins	`shockpack.margins` (and optional `dynamic_margins` for Yahoo/file/HTTP calibration — API.md)

II.8 Optional next steps

- **Catalogue-managed shock packs:** register a base spec, then call with `shockpack_catalog_entry_id` + optional inline `shockpack` patch — see API.md.
- **Portfolio:** `POST /v1/simulate/portfolio` for multiple correlated assets.
- **Codegen:** import `/openapi.json` into your stack (OpenAPI generators, Postman, etc.).

II.9 Other docs in this repo

Doc	Use
This file (`Azraq_Monte_Carlo_API_Reference.md`)	Versions, integration, flow, validation, full route reference in one place
ENDPOINTS.md	Same route catalogue as Part IV (kept for PDF script + short links)
API.md	Longer narrative, `response_help`, security notes, shared types
PROJECT_OVERVIEW.md	Product-level engine description
SCENARIO_LAB_NON_TECH_GUIDE.md	Scenario Lab UI (`/app/`)

Part III — Information flow, shocks, and validation

This section is for teams who can call the API but need a **clear picture of what travels where**, what **"shocks"** mean in this product, and **how to prove** that your own numbers actually move the results.

III.1 Two separate layers (this is the mental model)

Layer	JSON object	Role
Project economics	`asset` (`AssetAssumptions`)	*Your* deal: revenue, opex, capex, horizon, financing, covenant floor, optional transforms / full-stack flags. **No randomness here**—these are the baseline inputs the engine stresses.
Uncertainty model	`shockpack` (`ShockPackSpec`)	*How the world wobbles*: how many scenarios, seed, sampling method, **which risk factors** (`factor_order`), **how volatile** each is (`margins`), **how they move together** (`correlation` , `copula`), optional **paths over time** (`time_grid`), optional **calibration from files/HTTP/Yahoo** (`dynamic_margins`).

Monte Carlo = draw many correlated random **shock vectors** → for each draw, recompute coverage / cashflow-style metrics → summarise as distributions (percentiles, breach probability, etc.).

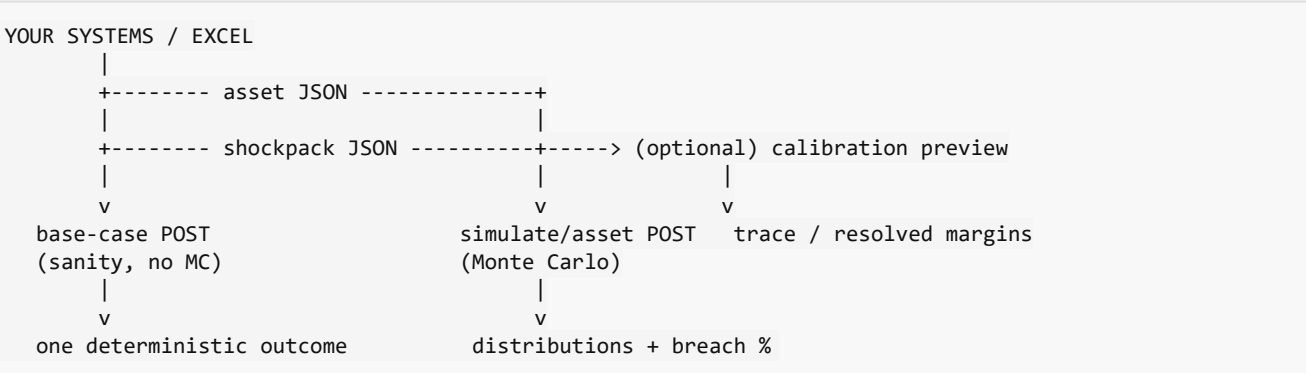
Shocks are not "events" you type in one-by-one (like "−20% revenue"). They are **parameters of random draws**: e.g. lognormal volatility on revenue (`revenue_log_sigma`) plus correlation with capex, opex, and rate. A **stress** in the colloquial sense is achieved by **changing those parameters** (wider sigmas, different correlation, or deterministic bumps via `factor_transforms` on the asset—see §III.4).

III.2 End-to-end flow (what to call, in order)

Start here — five steps (no diagram required)

1. Build **`asset`** (`AssetAssumptions`) from your spreadsheet or systems — the project as numbers you believe today.
2. Build **`shockpack`** (`ShockPackSpec`) — how many scenarios, seed, volatilities, correlation, optional **`dynamic_margins`** (Yahoo / file / HTTP).
3. **Optional:** `POST /v1/calibration/preview` with that shockpack — confirms external data filled **`margins`** without running MC (see **`calibration_trace`**).
4. **`POST /v1/simulate/v0/base-case`** with **`asset`** **only** — one deterministic row; must match your spreadsheet logic before you trust random runs.
5. **`POST /v1/simulate/asset`** with **`asset`** + **`shockpack`** — full Monte Carlo; read **`SimulationResult.metrics`** .

Same idea as a simple picture (read top → bottom)



Optional later (only if you need them)

- **Several projects together:** `POST /v1/simulate/portfolio` (2+ assets, one shared shock world).
- **Save and compare runs:** `POST /v1/snapshots/save` and `GET /v1/snapshots/diff/asset-metrics` .
- **Reuse a stored shockpack:** `POST /v1/shockpack/catalog/register` then call simulate with **`shockpack_catalog_entry_id`** .

- **Browse stress-source ideas:** `GET /v1/calibration/stress-data-catalog` ; **inspect** a series: `GET /v1/market/yfinance/{symbol}/history` (does not run MC by itself).

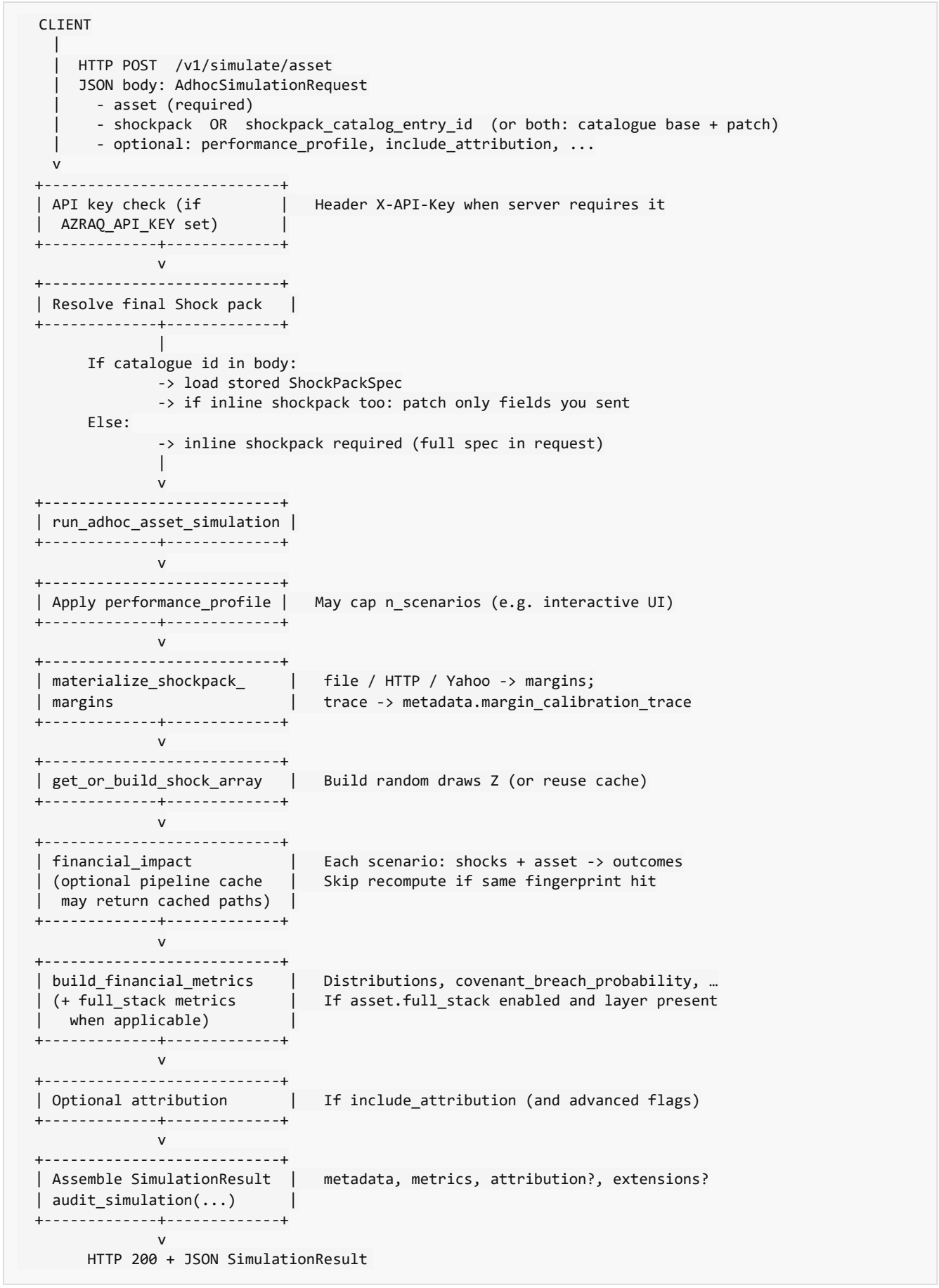
Endpoint one-liners

- `POST /v1/simulate/v0/base-case` : same `asset` shape as MC, **no** random draws — align with a spreadsheet first.
- `POST /v1/calibration/preview` : turns `dynamic_margins` into concrete `margins` ; no simulation. Use `calibration_trace` / run `metadata.margin_calibration_trace` to prove real data landed.
- `POST /v1/simulate/asset` : main MC — `asset` + `shockpack` → `SimulationResult` .

III.2a `POST /v1/simulate/asset` — full server flow (flowchart)

Who this is for: anyone who wants to see what happens **after** the HTTP POST, in one picture.

Flowchart (read top → bottom)



Explain the flowchart (plain language)

1. **You send one request** with the **project** (`asset`) and how to draw risk (`shockpack` and/or a **catalogue id** that points at a saved shock pack).
2. **Resolve** picks the **final shock recipe**: catalogue row first, then any fields you put inline **fill in or override** missing pieces. You cannot omit both catalogue and inline pack.
3. **Performance profile** can **reduce** scenario count for speed (`interactive` / `standard`); `deep` does not cap.
4. **Materialize margins** runs **before** random draws: if the shock pack asked for Yahoo/file/HTTP calibration, those numbers are merged into `margins` now. Check `metadata.margin_calibration_trace` to prove what happened.
5. **Shock array** builds (or loads from cache) the correlated random `Z` for all scenarios.
6. **Financial impact** walks every scenario through the model (or reuses cached outcomes when the pipeline cache hits).
7. **Metrics** turns raw paths into **percentiles**, **breach probability**, optional tail stats, optional **full_stack** block.
8. **Attribution** (only if requested) adds **"what drove the bad tail"** style diagnostics — indicative, not a bank sign-off.
9. **Response** is `SimulationResult` JSON; the server also **writes an audit row** for the run.

One line: `POST /v1/simulate/asset` = merge shock recipe → calibrate margins → draw shocks → value the asset many times → summarise → return JSON.

III.3 What happens inside one Monte Carlo run (simplified)

Relation to §III.2a: The numbered steps below are the **inside** of the `run_adhoc_asset_simulation` box in **§III.2a** (from "materialize margins" through "metrics"). §III.2a adds the **HTTP**, **resolve shock pack**, and **response/audit** wrapper.

This section is **one request on the server** — not the order you build JSON in your team.

One-line picture

```
shockpack  -->  (optional) fill margins from Yahoo / file / HTTP
                  |
                  v
            draw random Z  (many scenarios)
                  |
asset  -----------+------->  value each scenario (DSCR, IRR, …)
                  |
                  v
            pack into metrics (percentiles, breach %)
```

Steps (read 1 → 4)

1. **Lock in margins (optional calibration):** If `dynamic_margins` is set, the service **merges** file / HTTP / Yahoo into `margins` , then drops `dynamic_margins` from the resolved spec. Proof: `margin_calibration_trace` on the run when used.
2. **Draw random shocks:** Build **Z** (correlated normals / Student-t scaling), size from `n_scenarios` , `factor_order` , `correlation` , `copula` , optional `time_grid` .

3. **Map shocks to cash drivers:** Apply optional `asset.factor_transforms`, then lognormal / rate rules from `margins` so each scenario has stressed revenue, capex, opex, rate paths.
4. **Summarise:** Each scenario yields DSCR, IRR, etc.; the API returns percentiles and `covenant_breach_probability` in `metrics`.

Code names (only if you grep the repo): `materialize_shockpack_margins` → `get_or_build_shock_array` → `financial_impact`.

So: `margins` **control how wide** the distributions are; `correlation` controls **joint bad years**; `asset.*` controls **the size of the machine** being shocked.

III.4 Shock-related fields and what they do to outputs

Input (typical)	Effect on the run
`n_scenarios`	More scenarios → smoother percentiles; law of large numbers (diminishing noise).
`seed`	Reproducibility: same seed + same spec → same Z (subject to performance/cache notes in API docs).
`sampling_method`	`monte_carlo` vs `latin_hypercube` / `sobol` — changes **how** space is filled, not the marginal distributions.
`factor_order` + `correlation`	Must be square and same length. Higher positive correlation between revenue and opex (for example) moves **joint** tail risk.
`margins.revenue_log_sigma` (etc.)	**Wider** log σ → **wider** revenue (and downstream DSCR / IRR) distribution; breach probability often **rises** if downside matters for covenant.
`margins.rate_shock_sigma`	Volatility of **additive** shocks to the **annual** interest rate (decimal σ); increases rate path dispersion.
`copula` / `t_degrees_freedom`	Gaussian vs heavier tails on the dependence structure.
`time_grid`	Multi-period Z: e.g. min DSCR over life vs single-period snapshot (see API narrative).
`asset.factor_transforms`	**Deterministic** scaling of shock indices or level multipliers (e.g. stress capex level) — useful when you want a **known** wedge on top of stochastic draws.
`macro_regime`	**Label** for audit / catalogue; does not by itself change maths unless you pair it with different specs per regime.

External "real data" does **not** stream into every scenario. It is used to **set or update** the `margins` (and you verify that via **preview** or `margin_calibration_trace`).

III.5 Step-by-step: prove your data affects the outcome

Do these in order the first time you integrate.

Step A — Health and contract

1. `GET /health` — environment up.
2. Open `/openapi.json` or Part IV — authoritative list of fields and types.

Step B — Deterministic baseline (asset only)

1. `POST /v1/simulate/v0/base-case` with **only** your `asset` (same shape as in MC).
2. Check `base` metrics against your internal spreadsheet for the same inputs. If this fails, fix `asset` mapping before touching shocks.

Step C — One-dimensional sensitivity (asset)

For each critical field (`base_revenue_annual` , `initial_capex` , `financing.interest_rate_annual` , `financing.covenant_dscr` , …):

1. Change **one** field by a **small, known** amount (e.g. +5% revenue).
2. Re-run base-case; confirm the output moves in the **direction you expect** (e.g. higher revenue → better coverage if nothing else breaks validation).

This proves **your economic payload** is wired.

Step D — Shock width sensitivity (shockpack)

Fix `asset` and `seed` . Run `POST /v1/simulate/asset` twice:

1. **Low vol:** small `revenue_log_sigma` , `capex_log_sigma` , `opex_log_sigma` , `rate_shock_sigma` .
2. **High vol:** multiply those sigmas by e.g. **2×** (keep correlation identical).

Expect: breach probability and dispersion of DSCR / IRR **increase** with higher σ. If nothing moves, you are likely not hitting the simulate route you think you are, or metrics are cached from an identical fingerprint—compare `metadata.run_id` and request bodies.

Step E — Calibration path (optional)

1. Put a minimal margin patch in JSON or Excel; reference it in `dynamic_margins.file` , or configure `dynamic_margins.yahoo_finance` with a liquid symbol.
2. `POST /v1/calibration/preview` with the same `shockpack` you intend to use.
3. Read `resolved_shockpack.margins` and `calibration_trace` . Confirm numbers match your file / Yahoo-derived expectation.
4. Run `POST /v1/simulate/asset` and confirm `metadata.margin_calibration_trace` is populated and consistent with preview.

Step F — Portfolio (if applicable)

1. Two assets, shared `shockpack` , `POST /v1/simulate/portfolio` .
2. Toggle correlation between a macro factor shared across assets (via `correlation`) and confirm **portfolio** breach / joint tail metrics respond.

III.6 Short “video” outline (e.g. 8-minute screen recording)

If you record a walkthrough (Loom, Teams, etc.), this sequence matches **§III.2** above and builds confidence:

1. **0:00–0:45** — Show `/docs` + `openapi.json` ; say “two blocks: `asset` + `shockpack` ”.
2. **0:45–2:30** — Base-case POST: paste minimal `asset` , show response vs a spreadsheet row.
3. **2:30–4:30** — Same `asset` , two MC runs: low vs high `*_log_sigma` ; show `covenant_breach_probability` and DSCR percentiles.
4. **4:30–6:30** — Calibration preview: `dynamic_margins` with a tiny JSON file or one Yahoo binding; show resolved margins + trace.
5. **6:30–8:00** — Full simulate with trace; point to `response_help` in JSON for non-technical readers.

III.7 One-line summary for stakeholders

You supply a deterministic project model (`asset`) and a stochastic specification (`shockpack`); the API returns distributions of financial outcomes. External series are optional helpers to set volatility parameters—verify them with calibration preview and `margin_calibration_trace` , then validate behaviour with base-case checks and controlled σ sensitivity.

Part IV — API endpoints (full reference)

Hosted (Swagger UI — try requests in the browser): https://monte-carlo.octaloop.dev/docs#/

Same deployment: OpenAPI JSON — `https://monte-carlo.octaloop.dev/openapi.json` · optional investor dashboard — `https://monte-carlo.octaloop.dev/app/` (if enabled).

Local dev: `http://127.0.0.1:8000` (same paths: `/docs` , `/openapi.json` , `/app/`).

Machine-readable contract: `GET /openapi.json` . **Interactive UI:** `GET /docs` (Swagger).

Shorter route-only catalogue: **ENDPOINTS.md** (kept for diff-friendly edits; this file is the full narrative + reference).

IV.0 Contents (Part IV)

IV.0 Complete route inventory

Application API (implemented in `azraq_mc/api.py`)

There are **20 HTTP operations** and **1 WebSocket** — **21** interfaces total.

#	Method	Path	Summary
1	`GET`	`/health`	Liveness (no API key).
2	`GET`	`/v1/market/yfinance/{symbol}/history`	Yahoo OHLCV (query `period`).
3	`GET`	`/v1/market/yfinance/{symbol}/returns`	Yahoo simple returns.
4	`GET`	`/v1/market/ecb/eur-usd`	ECB EUR/USD spot.
5	`GET`	`/v1/calibration/stress-data-catalog`	Stress-source catalogue.
6	`GET`	`/v1/catalog/full-stack-factors`	12-factor full-stack list.
7	`POST`	`/v1/calibration/preview`	Resolve `dynamic_margins` → `margins` (no MC).
8	`POST`	`/v1/simulate/scheduled/asset`	MC + optional snapshot file.

#	Method	Path	Summary
9	`POST`	`/v1/simulate/asset`	**Main** single-asset Monte Carlo.
10	`POST`	`/v1/simulate/v0/base-case`	Deterministic baseline (body = `AssetAssumptions`).
11	`POST`	`/v1/simulate/portfolio`	Joint MC, ≥2 assets.
12	`POST`	`/v1/shockpack/catalog/register`	Save shock pack → `entry_id`.
13	`POST`	`/v1/shockpack/catalog/{entry_id}/promote`	Change `promotion_tier`.
14	`GET`	`/v1/shockpack/catalog/{entry_id}`	Fetch catalogue row + `spec`.
15	`GET`	`/v1/shockpack/catalog`	List catalogue entries.
16	`POST`	`/v1/shockpack/export/npz`	Write shock `Z` to `.npz` on disk.
17	`POST`	`/v1/snapshots/save`	Save prior result JSON.
18	`GET`	`/v1/snapshots/list`	List snapshot paths.
19	`GET`	`/v1/audit/runs`	Recent audit rows (`limit`).
20	`POST`	`/v1/snapshots/diff/asset-metrics`	Diff two asset `SimulationResult` files.
21	`WebSocket`	`/ws/v1/simulate/portfolio`	Portfolio MC + optional progress.

Anything not in this table is **not** a custom business route (see below).

Also served (FastAPI + static UI)

These are automatic or mounted by the app, not separate “risk” APIs:

Kind	Path	Purpose
OpenAPI schema	`GET /openapi.json`	Machine-readable spec (codegen, Postman).
Swagger UI	`GET /docs`	Try-it-out browser UI.
ReDoc	`GET /redoc`	Alternate docs UI (if enabled by FastAPI defaults).
Static dashboard	`GET /app/…`	Investor UI (`frontend/`), only if that folder exists.

IV.1 Authentication & headers

Condition	Client behaviour
Server has `AZRAQ_API_KEY` set	Send header `X-API-Key: <same value>` on all routes **except** `GET /health` (health is public).
Optional audit	`X-Azraq-User-Id` — label stored with runs (not authentication).
Catalogue / tenant	`X-Azraq-Tenant-Id` — filters or enforces tenant on some **catalog** `GET` routes.
Promoting shock packs	`POST /v1/shockpack/catalog/{entry_id}/promote` needs `X-Azraq-Catalog-Role` in the allow-list (`AZRAQ_CATALOG_PROMOTER_ROLES`), unless that env is empty (local dev).

WebSocket: if an API key is required, pass header `x-api-key` on the handshake (see portfolio WS below).

IV.2 Shared JSON models

These names match **Pydantic / OpenAPI** schemas in the repo. Field-level detail: `/docs` or API.md § "Shared types".

Model	Used for
`FinancingAssumptions`	`debt_principal`, `interest_rate_annual` (decimal), `loan_term_years`, `covenant_dscr`
`AssetAssumptions`	Project economics + `financing` + optional `utility_opex_annual`, `factor_transforms`, `full_stack`, …
`ShockPackSpec`	`shockpack_id`, `seed`, `n_scenarios`, `sampling_method`, `margins`, optional `dynamic_margins`, `correlation`, `factor_order`, …
`AdhocSimulationRequest`	`asset` + `shockpack` and/or `shockpack_catalog_entry_id` + optional attribution / `performance_profile`
`ScheduledAssetRequest`	Same as adhoc + `label`, `persist`, `model_version`
`PortfolioSimulationRequest`	`portfolio_id`, `portfolio_assumption_set_id`, `assets` (array, ≥ **2**), `shockpack` or catalogue id
`SimulationResult`	`metadata`, `metrics`, optional `attribution`, `full_stack`, `extensions`
`BaseCaseResult`	Deterministic run: `metadata`, `base` (point estimates)
`PortfolioSimulationResult`	`metadata`, `per_asset[]`, `portfolio`
`CalibrationPreviewResponse`	`resolved_shockpack`, `calibration_trace`
`ScheduledAssetResponse`	`result` (`SimulationResult`), `snapshot_path`

Most JSON responses also include `response_help` (plain-language hints).

IV.3 How to send data and read results

Single-project Monte Carlo (typical integration)

1. **Build** an `AssetAssumptions` object from your systems (revenue, opex, capex, debt, rates, covenant, horizon, ids).
2. **Build** a `ShockPackSpec` (at least `shockpack_id`, `seed`, `n_scenarios`, `sampling_method`, and usually `margins` or `dynamic_margins` for calibration).
3. **POST** `/v1/simulate/asset` with body `{ "asset": {...}, "shockpack": {...}, ... }`.
4. **Read** `200` JSON `SimulationResult`: use `metrics.dscr`, `metrics.irr_annual`, `metrics.covenant_breach_probability`, tail fields, etc., and `metadata` for `run_id`, timing, seed.

State: there is **no session**. Each POST is independent. To "reset," send a new body.

Optional: validate calibration only (no MC)

POST `/v1/calibration/preview` with a full `ShockPackSpec` (e.g. with `dynamic_margins.yahoo_finance`). Response gives `resolved_shockpack` with concrete `margins` — paste that into simulate.

Optional: save results for diffing

- **POST** `/v1/simulate/scheduled/asset` with `persist: true` → get `snapshot_path`.
- Or **POST** `/v1/snapshots/save` with a prior `SimulationResult` in `result`.

Then **POST** `/v1/snapshots/diff/asset-metrics` with `before_path` / `after_path`.

IV.4 Endpoint reference

Legend: **Path** / **Query** / **Body** = where parameters go. **Auth** = needs `X-API-Key` when server key is set (except `/health`).

`GET /health`

Purpose	Liveness: server responds.
Parameters	None.
Response	`{ "status": "ok", "response_help": { ... } }`
Usage	Monitoring or pre-flight check. **No model run.**
Auth	None.

`GET /v1/market/yfinance/{symbol}/history`

Purpose	Raw OHLCV history from Yahoo (inspection / reporting).
Path	`symbol` — ticker, e.g. `HG=F`.
Query	`period` — `1d`, `5d`, `1mo`, `3mo`, `6mo`, `1y`, `2y`, `5y`, `10y`, `ytd`, `max` (default `1y`).
Response	`{ "symbol", "period", "data": [ ... ] }` + `response_help`. `404` if no data; `501` if `yfinance` missing.
Usage	Does **not** run Monte Carlo. For feeding vol into the engine use `dynamic_margins.yahoo_finance` or `/v1/calibration/preview`.
Auth	Optional key.

`GET /v1/market/yfinance/{symbol}/returns`

Purpose	Close-to-close **simple** returns series (human review).
Path	`symbol`.
Query	`period` — same set as history.
Response	`{ "symbol", "period", "returns": [{"date","return"},...], "n" }` + `response_help`. `404` / `501` as above.
Usage	Exploratory; internal calibration uses **log** returns + annualisation.
Auth	Optional key.

`GET /v1/market/ecb/eur-usd`

Purpose	ECB daily **EUR/USD** reference from public XML.
Parameters	None.
Response	Rate fields + `response_help`. `502` if feed fails.

Usage	FX sanity checks; wire into `dynamic_margins` via your own JSON/file/http if needed.
Auth	Optional key.

`GET /v1/calibration/stress-data-catalog`

Purpose	Catalogue of stress / data sources and integration hints (Appendix J–style).
Parameters	None.
Response	`{ "sources", "integration_overview", "response_help" }`.
Usage	Discover URLs and how they map to `dynamic_margins`.
Auth	Optional key.

`GET /v1/catalog/full-stack-factors`

Purpose	Ordered `factor_order` + metadata for **12-factor full-stack** mode.
Parameters	None.
Response	`{ "factor_order", "risk_factors", "response_help" }`.
Usage	Required before `asset.full_stack` + matching 12-factor `ShockPackSpec`.
Auth	Optional key.

`POST /v1/calibration/preview`

Purpose	Resolve `dynamic_margins` (file / http / Yahoo) into numeric `margins` **without** drawing scenarios.
Body	`ShockPackSpec` (JSON) — same shape as `shockpack` on simulate.
Response	`CalibrationPreviewResponse`: `resolved_shockpack`, `calibration_trace`.
Usage	Governance: confirm `resolved_shockpack.margins` before expensive `/v1/simulate/asset`.
Auth	Optional key.

`POST /v1/simulate/v0/base-case`

Purpose	**Deterministic** single path (no random shocks). Fast tie-out vs spreadsheet.
Body	`AssetAssumptions` as **root** JSON (not wrapped in `{ "asset": ... }`).
Response	`BaseCaseResult`: `metadata` (`execution_mode: "v0_base"`), `base` point estimates (`dscr`, `irr_annual`, …).
Usage	Baseline only — **not** tail risk.
Auth	Optional key.

`POST /v1/simulate/asset`

Purpose	Main **Monte Carlo** for **one** asset — DSCR/IRR distributions, breach probability, etc.

Body	`AdhocSimulationRequest` : `asset` (required); `shockpack` and/or `shockpack_catalog_entry_id` ; optional `include_attribution` , `include_advanced_attribution` , `attribution_tail_fraction` , `performance_profile` .
Response	`SimulationResult` : `metadata` , `metrics` , optional `attribution` , `full_stack` , `extensions` .
Usage	Primary integration: **send** economics + shock spec, **receive** `metrics` for dashboards or storage.
Auth	Optional key.

`POST /v1/simulate/scheduled/asset`

Purpose	Same maths as `/v1/simulate/asset` , optional **on-disk JSON snapshot** for monitoring / diff.
Body	`ScheduledAssetRequest` : all `AdhocSimulationRequest` fields + `label` , `persist` (default true), `model_version` .
Response	`ScheduledAssetResponse` : `result` (`SimulationResult`), `snapshot_path` or `null` .
Usage	Recurring runs; pair with `/v1/snapshots/diff/asset-metrics` . Snapshot root: `AZRAQ_SNAPSHOT_DIR` or default `data/snapshots` .
Auth	Optional key.

`POST /v1/simulate/portfolio`

Purpose	**Joint** Monte Carlo for **≥ 2** assets (same shock draw per scenario across names).
Body	`PortfolioSimulationRequest` : `shockpack` or catalogue id; `portfolio_id` , `portfolio_assumption_set_id` , `assets` (array length ≥ 2); optional `performance_profile` .
Response	`PortfolioSimulationResult` : `metadata` (includes `factor_order` , `factor_correlation` , `copula` from the resolved shock pack), `per_asset[]` (same shape as single-asset `metrics`), `portfolio` — jointly computed pool metrics (**§IV.6.3.1**).
Usage	Portfolio concentration; joint breach ≠ sum of silo runs; compare weakest-link vs blended vs max DSCR paths and blend IRR tails on `portfolio` .
Auth	Optional key.

`WebSocket` `WS /ws/v1/simulate/portfolio`

Purpose	Same as `POST /v1/simulate/portfolio` with optional **progress** messages.
Handshake	If `AZRAQ_API_KEY` set: header `x-api-key` .
Messages	Client sends **one** JSON = `PortfolioSimulationRequest` . Server: optional `{ "type":"progress", ... }` , then `{ "type":"result", "body": <PortfolioSimulationResult> }` or `{ "type":"error", "detail" }` .
Usage	Long runs from browser / clients that prefer streaming.
Auth	`x-api-key` when server key is set.

`POST /v1/shockpack/catalog/register`

Purpose	Store a `ShockPackSpec` in the catalogue for reuse.

Body	`ShockPackSpec` (JSON).
Query	`semver` (default `1.0.0`), optional `tenant_id`, `promotion_tier` (`dev` / `staging` / `prod`), `rbac_owner_role`.
Response	`{ "entry_id": "<uuid>", "response_help": ... }` — use `entry_id` as `shockpack_catalog_entry_id` on simulate.
Usage	Freeze governance-approved shock design; inline `shockpack` can still patch fields.
Auth	Optional key.

`POST /v1/shockpack/catalog/{entry_id}/promote`

Purpose	Change catalogue `promotion_tier` (e.g. dev → prod).
Path	`entry_id`.
Query	`to_tier` — `dev` | `staging` | `prod`.
Headers	`X-Azraq-Catalog-Role` unless promoter env list is empty.
Response	`{ "ok": true, "response_help": ... }` — `403` if role not allowed.
Usage	Gate “approved” packs in `GET /v1/shockpack/catalog?promotion_tier=prod`.
Auth	Optional key + promoter role.

`GET /v1/shockpack/catalog/{entry_id}`

Purpose	Fetch one catalogue row including full `spec`.
Path	`entry_id`.
Headers	Optional `X-Azraq-Tenant-Id` — must match entry tenant if enforced.
Response	Catalogue object + `response_help`. `403` tenant mismatch; `404` missing.
Usage	Inspect exact shock JSON before production runs.
Auth	Optional key.

`GET /v1/shockpack/catalog`

Purpose	List recent registrations (**metadata**; use **GET by id** for full `spec`).
Query	`limit` (default 100), optional `tenant_id`, `promotion_tier`. Header tenant may filter.
Response	`{ "entries": [ ... ], "response_help": ... }`.
Usage	Discover `entry_id` for CI/CD.
Auth	Optional key.

`POST /v1/shockpack/export/npz`

Purpose	Write correlated shock array `Z` to a `.npz` file on the server (offline HPC / custom models).

Body	`ShockExportRequest` : `shockpack` (required), optional `directory` .
Response	`{ "path": "<absolute .npz path>", "response_help": ... }` . `dynamic_margins` resolved before draw.
Usage	No DSCR/IRR — only random draws. Default dir: `AZRAQ_SHOCK_EXPORT_DIR` or `data/shockpacks` .
Auth	Optional key.

`POST /v1/snapshots/save`

Purpose	Persist an **already computed** result to JSON on disk.
Body	`SnapshotSaveRequest` : optional `label` , `result` = full `SimulationResult` / `PortfolioSimulationResult` / `BaseCaseResult` .
Response	`{ "path": "<absolute path>", "response_help": ... }` .
Usage	Bookmark or feed external tools; same root as scheduled snapshots.
Auth	Optional key.

`GET /v1/snapshots/list`

Purpose	List snapshot **file paths** under the snapshot root.
Parameters	None.
Response	`{ "paths": [ "..." ], "response_help": ... }` .
Usage	Pick `before_path` / `after_path` for diff.
Auth	Optional key.

`POST /v1/snapshots/diff/asset-metrics`

Purpose	Compare two **saved** asset `SimulationResult` snapshots.
Body	`{ "before_path": "<server path>", "after_path": "<server path>" }` .
Response	`metrics_delta` , `provenance_delta` , `response_help` .
Usage	Explain drift: assumptions vs shock vs seed vs model version. `400` if either file is not an asset MC result.
Auth	Optional key.

`GET /v1/audit/runs`

Purpose	Recent simulation events from SQLite audit DB (if enabled).
Query	`limit` (default 50).
Response	`{ "runs": [ { "run_id", "run_kind", "asset_id", "shockpack_id", "seed", "n_scenarios", ... } ], "response_help": ... }` .
Usage	Correlate `run_id` with `SimulationResult.metadata.run_id` . Empty if audit disabled/missing.
Auth	Optional key.

IV.5 Errors

HTTP	Typical cause
401	Missing or wrong `X-API-Key` when server requires it.
403	Catalogue tenant / promoter role mismatch.
404	Yahoo history missing; unknown catalogue `entry_id`.
400	Validation: bad JSON shape, wrong asset count for portfolio, wrong snapshot types for diff.
422	Pydantic validation detail (FastAPI) — check `detail` array for field paths.
501	Optional dependency missing (e.g. `yfinance`).
502	Upstream feed failure (e.g. ECB).

For `422`, use `/docs` "Try it out" or client validation against `/openapi.json` to align payloads before production.

IV.6 Live deployment, examples, and response detail

IV.6.1 Public host (Octaloop)

Resource	URL
Swagger UI (schemas + "Try it out")	https://monte-carlo.octaloop.dev/docs#/
OpenAPI JSON (import to Postman, codegen)	`https://monte-carlo.octaloop.dev/openapi.json`
Web app (if deployed)	`https://monte-carlo.octaloop.dev/app/`

Use `https://monte-carlo.octaloop.dev` as the host for the paths in §IV.4 (e.g. `POST https://monte-carlo.octaloop.dev/v1/simulate/asset`). If the server has `AZRAQ_API_KEY` set, add header `X-API-Key` on calls other than `GET /health`.

IV.6.2 `AdhocSimulationRequest` — request fields (main Monte Carlo)

Field	Required	Type / notes
`asset`	Yes	Full `AssetAssumptions` (ids, horizon, revenue, opex, capex, equity, financing, optional `utility_opex_annual`, `factor_transforms`, `full_stack`, …).
`shockpack`	One of pack or catalogue	Inline `ShockPackSpec` (`shockpack_id`, `seed`, `n_scenarios`, `sampling_method`, `margins`, optional `dynamic_margins`, `correlation`, …).
`shockpack_catalog_entry_id`	One of pack or catalogue	UUID from `POST /v1/shockpack/catalog/register`; inline `shockpack` can patch fields.
`include_attribution`	No	Default `false`. Tail attribution (indicative narrative).
`include_advanced_attribution`	No	Default `false`. Extra factor weights when attribution on.
`attribution_tail_fraction`	No	Default `0.05` (e.g. worst ~5% scenarios for attribution).
`performance_profile`	No	`"interactive"` (caps `n_scenarios`, good for UI), `"standard"`, `"deep"` (no cap), or `null`.

IV.6.3 `SimulationResult.metrics` — main outputs (read after `POST /v1/simulate/asset`)

Field	Meaning
`dscr`	`DistributionSummary`: percentiles (e.g. `p05` – `p95`), `mean`, `std` — debt service coverage over the run.
`irr_annual`	`DistributionSummary` for equity IRR (decimal per year, e.g. `0.08` = 8%).
`total_capex`	Optional `DistributionSummary` for stochastic nominal build cost when the engine supplies it.
`covenant_breach_probability`	Share of paths where modeled DSCR falls below `asset.financing.covenant_dscr`.
`probability_of_default_proxy_dscr_lt_1`	PD-style proxy using DSCR < 1 (see OpenAPI / product definition).
`var_irr_95` / `cvar_irr_95`	Tail / shortfall style metrics on IRR when populated.
`ebitda`, `levered_cf`, `nav_proxy_equity`, …	Other `DistributionSummary` blocks when enabled by model path.

Always read `metadata` for `run_id`, `seed`, `n_scenarios`, `compute_time_ms`, `shockpack_id`, optional `margin_calibration_trace` (if Yahoo/file/http calibration ran).

IV.6.3.1 `PortfolioSimulationResult` — `portfolio` object (`PortfolioMetrics`)

`per_asset[i].metrics` is the same family as `POST /v1/simulate/asset`. `portfolio` adds **whole-book views** derived from **the same Monte Carlo paths** (one shock draw index shared across assets).

Field	Meaning
`probability_any_covenant_breach`	Share of scenarios where **at least one** asset breaches its covenant.
`probability_at_least_k_breaches`	`"2"`, `"3"`, … for **multiple simultaneous** breaches (`k` ≥ 2).
`min_dscr_across_assets`	`DistributionSummary`: each scenario uses the **minimum** path DSCR across assets (weakest-link view).
`max_dscr_across_assets`	`DistributionSummary`: each scenario uses the **maximum** path DSCR across assets (strongest name).
`revenue_weighted_mean_dscr_across_assets`	`DistributionSummary`: each scenario $\Sigma\ w_i \cdot DSCR_i$ with $w_i \propto$ `base_revenue_annual` — **blend** at the portfolio level (**not** a merged waterfall statement). Same weights underlie `weighted_covenant_breach_exposure` / `revenue_herfindahl`.
`revenue_weighted_mean_equity_irr_across_assets`	`DistributionSummary`: revenue-weighted mean equity IRR paths (blend; **not** consolidated-fund IRR from pooled cashflows).
`var_irr_95` / `cvar_irr_95`	**Same conventions** as `SimulationResult.metrics`: IRR tail shortfall (**median − p05**) and **mean IRR in worst ~5%** on the **revenue-weighted blend IRR** series.
`sum_levered_cf_year1`	`DistributionSummary` of summed **year-1** after-tax levered CF across assets.
`var_sum_levered_cf_p05` / `cvar_sum_levered_cf_p05`	Raw **p05** summed CF level and mean CF in scenarios at/below it (portfolio cashflow downside).

Field	Meaning
`revenue_herfindahl`	Σw_i^2 concentration on revenue weights w_i.
`weighted_covenant_breach_exposure`	$E[\Sigma w_i \cdot \mathbb{1}(\text{breach}_i)]$ — revenue-weighted covenant stress simultaneously across names.
`cross_asset_dscr_correlation_pearson`	Symmetric Pearson **ρ** of **scenario-by-scenario DSCR paths** (row/column order matches `metadata.asset_ids`). Output co-movement, **not** the 4×4 factor `correlation` input matrix.
`cross_asset_equity_irr_correlation_pearson`	Same for **IRR paths** (order matches `metadata.asset_ids`).

Macro **factor ρ** assumed for shocks appears on `PortfolioRunMetadata` as `factor_order`, `factor_correlation`, `copula`.

IV.6.4 Example: health check (no API key)

Replace `HOST` with `https://monte-carlo.octaloop.dev` or `http://127.0.0.1:8000`.

```
curl -sS "${HOST}/health"
```

IV.6.5 Example: single-asset Monte Carlo (minimal body)

Send `Content-Type: application/json`. If an API key is required, add `-H "X-API-Key: YOUR_KEY"`.

```
curl -sS -X POST "${HOST}/v1/simulate/asset" \
  -H "Content-Type: application/json" \
  -d '{
    "shockpack": {
      "shockpack_id": "demo-sp-1",
      "seed": 42,
      "n_scenarios": 800,
      "sampling_method": "monte_carlo",
      "margins": {
        "revenue_log_mean": 0,
        "revenue_log_sigma": 0.08,
        "capex_log_mean": 0,
        "capex_log_sigma": 0.06,
        "opex_log_mean": 0,
        "opex_log_sigma": 0.05,
        "rate_shock_sigma": 0.005
      }
    },
    "asset": {
      "asset_id": "demo-asset-1",
      "assumption_set_id": "demo-as-1",
      "horizon_years": 8,
      "base_revenue_annual": 12000000,
      "base_opex_annual": 5000000,
      "initial_capex": 80000000,
      "equity_fraction": 0.35,
      "tax_rate": 0.0,
      "financing": {
        "debt_principal": 40000000,
        "interest_rate_annual": 0.055,
        "loan_term_years": 12,
        "covenant_dscr": 1.2
      }
    },
    "include_attribution": false,
    "performance_profile": "interactive"
  }'
```

Note: The JSON above uses a fully specified `margins` block. If `dynamic_margins` (Yahoo/file/http) is used instead, run `POST /v1/calibration/preview` first to validate resolved sigmas. Omit `performance_profile` or use `"standard"` / `"deep"` for higher scenario limits (see OpenAPI).

IV.6.6 Example: deterministic baseline (no Monte Carlo)

Body is `AssetAssumptions` only (not wrapped in `asset`).

```
curl -sS -X POST "${HOST}/v1/simulate/v0/base-case" \
  -H "Content-Type: application/json" \
  -d '{"asset_id":"demo","assumption_set_id":"v0","horizon_years":8,
       "base_revenue_annual":12000000,"base_opex_annual":5000000,
       "initial_capex":80000000,"equity_fraction":0.35,"tax_rate":0,
       "financing":{"debt_principal":40000000,"interest_rate_annual":0.055,
       "loan_term_years":12,"covenant_dscr":1.2}}'
```

IV.6.7 WebSocket portfolio run

Connect to `wss://monte-carlo.octaloop.dev/ws/v1/simulate/portfolio` (or `ws://127.0.0.1:8000/...` locally). Send **one** message: same JSON as `POST /v1/simulate/portfolio`. If API key is set, send header `x-api-key` on the handshake.

IV.6.8 Where to see every field

Open **Swagger UI**, expand `POST /v1/simulate/asset`, inspect **Request body** and **Response** schemas (`AdhocSimulationRequest`, `SimulationResult`, `FinancialRiskMetrics`, …). The live schema explorer is authoritative for nested optional fields.

Related reference files (repo)

File	Role
API.md	Full narrative, examples, `response_help`, security notes, shared types
ENDPOINTS.md	Shorter endpoint catalogue
INTEGRATION.md	Short integration guide (also Part II in this file)
MONTE_CARLO_FLOW_AND_VALIDATION.md	Flow-only doc (also Part III in this file)
`Azraq_Monte_Carlo_API_Reference.pdf`	Print-ready PDF from `python scripts/build_api_reference_pdf.py`

End of complete guide.